

U.S.N.

B.M.S. College of Engineering, Bengaluru-560019

Autonomous Institute Affiliated to VTU

June 2025 Semester End Main Examinations**Programme: B.E.****Semester: VI****Branch: AI & DS/CSE(DS)/CSE(IOT)****Duration: 3 hrs.****Course Code: 23DC6PCSEA****Max Marks: 100****Course: Software Engineering & Agile Methodologies**

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
2. Missing data, if any, may be suitably assumed.

Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.			UNIT - I	CO	PO	Marks
	1	a)	Discuss the needs of layered technology in software engineering. Explain SDLC with neat diagram.	CO1	PO1	08
		b)	Identify the fundamental activities of software engineering and agile development techniques.	CO1	PO1	06
		c)	Analyze the necessities and issues of software life cycle model for virtual doctor health recommendation problem.	CO2	PO2	06
			OR			
	2	a)	Construct waterfall model with neat diagram. Examine the problems encountered when applying the waterfall model.	CO1	PO1	08
		b)	Discuss the steps involved in rapid application development model with neat diagram.	CO1	PO1	06
		c)	Compare and contrast different life cycle models.	CO2	PO2	06
			UNIT - II			
	3	a)	Explain the SRS document and design document. Compile the skeletal structure of SRS document with the standard specified by IEEE/ANSI 830_1998.	CO2	PO2	06
		b)	Construct a Level-1 DFD for library management system software.	CO3	PO3	07
		c)	Perceive the types of requirements for "Online railway reservation system" with the help of requirements engineering and check the completeness and consistency of your requirements.	CO2	PO2	07
			OR			
	4	a)	Design and develop State Transition diagrams for coffee machine.	CO3	PO3	06

	b)	Compare and contrast cohesion and coupling in the context of software design.	CO2	PO2	07
	c)	Discover the difficulties and challenges in “customer interaction” process. Explain a formal procedure used for customer interaction.	CO2	PO2	07
		UNIT - III			
5	a)	Discuss the software project management concepts in detail.	CO1	PO1	06
	b)	Analyze any two metrics which are used to measure the software in detail. Discuss clearly the advantages and disadvantages of these metrics.	CO2	PO2	07
	c)	Discuss briefly on Empirical Estimation Techniques and how do you estimate the cost involved.	CO2	PO2	07
		OR			
6	a)	Discuss the COCOMO model with the help of an example.	CO1	PO1	06
	b)	Analyze the need for project scheduling and team structures in software project management and describe them with suitable example.	CO2	PO2	07
	c)	Make use of the values of size in KLOC and different cost drivers for a project are given below: size=200 KLOC Cost driver: Software reliability = 1.15 Use of software tools = 0.91 Product complexity = 0.85 Execution time constraint = 1.00 Calculate the effort for three types of projects viz. organic, semidetached and embedded, using COCOMO model.	CO2	PO2	07
		UNIT - IV			
7	a)	Explain the principles and elements of agile methods.	CO1	PO1	06
	b)	Considering a rapidly changing client requirement scenario, design and demonstrate how Agile provides better outcomes than traditional models like Waterfall in terms of project adaptability and stakeholder satisfaction.	CO3	PO3	07
	c)	Design the Kanban method to improve the workflow efficiency in a software development team, and discuss the potential limitations need to be considered when scaling it for larger projects.	CO3	PO3	07
		OR			
8	a)	Describe how using project boards and enriching issues within Kanban boards can enhance team collaboration and task visibility especially in managing complex software development projects.	CO1	PO1	06

		b)	In the context of Jira Scrum projects, design the Quick Search and Basic Search functionalities that can effectively be used to improve sprint planning and issue tracking efficiency.	CO3	PO3	07
		c)	Demonstrate the use of JQL to identify bottlenecks in a Scrum or Kanban project. Discuss the advantages it can offer over basic filtering methods in Jira.	CO3	PO3	07
			UNIT - V			
	9	a)	Construct an Extreme Programming (XP) life cycle that support continuous feedback and improvement. Analyse in what types of projects might this approach face limitations.	CO3	PO3	06
		b)	Formulate the structure and roles within an XP team that foster collaboration and high-quality code. Predict the possible challenges that could arise in a distributed team setting.	CO3	PO3	07
		c)	Analyze a class diagram for an SCM system with classes like Configuration Item, Version, Change Request, and Repository. Evaluate its suitability for managing version control and change management.	CO3	PO3	07
			OR			
	10	a)	Design a configuration management process that ensures consistency and traceability in large-scale software projects. Predict the risks can arise if this process is poorly implemented.	CO3	PO3	06
		b)	Develop a version control system contributing to collaborative software development. Discuss the challenges might teams face when integrating VCS into continuous integration workflows.	CO3	PO3	07
		c)	A Software Developer is trying to develop highly reliable software used in online shopping applications. He has to carry out thorough testing of the software in order to have negligible errors. Design the same by applying different tasks of SCM process.	CO3	PO3	07
